Current GPU model — streamed storage + throughput instrumentation
N20K-SYS-0.1 — host, ordered execution, shared SDRAM and independent scanout

Host CPU / driver — outside simulated GPU execution time

Transform and homogeneous clipping
Projection / Q4 triangle setup / lighting
Stable bins for 300 logical 32×32 tiles

prepared commands and pixels

compile_pass() / CPU packet encoder
Images + 320-byte prepared records
Tile table + ordered address lists
Typical stream: WRITE_HEAP, SUBMIT_PASS,
then separate FENCE and PRESENT commands

serialized request bytes

Current device model — one common 100.8 MHz core / SDR clock (configured assumption)

Clocked host interface — no direct C++ pass injection

Shared 8-bit parallel bus
25.2 MHz derived transfer cadence
≤1 byte / 4 core edges, one direction
Setup 1 / hold 1 core cycle
Register select + valid/read/ready

accepted write byte

RX FIFO: 256 bytes
1 × SDPB 1024×16
Low byte per native word

registered read / capture

Byte parser / protocol checks
16-byte NGPU header, version 0.1
Length ≤1024, opcode and sequence
Partial packets never execute

write staged payload
publish only complete packet

3 × 1024-byte packet banks
3 × SDPB 1024×16
2 published command slots
INCLUDING the active front lease
+ 1 decoder-owned staging bank

front-bank lease

Synchronous payload read port
Executor holds front-bank ownership
until that command retires

fresh packet boundary read selector
out-of-band controls
snapshot / capability byte
reply-byte read

Reply construction / serializer
1 × SDPB 1024×16
1 byte write / core edge
Separate synchronous read / capture
Reply: sequence + epoch + token

registered reply byte

TX reply FIFO: 1024 bytes
1 × SDPB 1024×16
Registered RAM + byte prefetch
Host pauses propagate backpressure

Sideband reads bypass command queue
Regs 1–63: coherent status snapshot
Regs 64–127: immutable capabilities
Status remains readable during faults

Out-of-band abort / reset
Abort: quiesce, drain issued memory,
flush queues, new epoch
Reset: restart initialization

latched request fields

cancel / drain / recover
established status / GET_CAPS / GET_STATUS reply

Ordered executor — exactly one command operation active; branches are alternatives

SystemModel executor FSM
Capture payload through packet RAM
Validate ranges / attachment ownership
Select one operation; retain its lease

opcode 5
opcode 3
opcode 4
address + length + data
completion
completion after swap

COPY_RECT / heap-image copy
Outside a render pass
Full clipped row, ≤640 RGB565 pixels
Reuse texture-cache bank as scratch
Row direction preserves overlap

WRITE_HEAP
Writer: one 32-bit pack + 4-bit mask
Source ≤1 byte / core edge
Uploads cannot write attachments

SUBMIT_PASS
Fetch descriptor, table, refs and records
One resident 32×32 tile
Load / draw / drain / store all 300 tiles
3 tile banks + shared texture bank

opcode 7
opcode 6
opcode 8
reply published to TX
next complete command

PRESENT
Seal initialized writable target
Wait for eligible vertical blank
Swap front; then publish completion

destination writes acknowledged
write acknowledgements

FENCE
Execute after earlier stores complete
Publish host token / sequence / epoch

texture owner

READ_MEMORY
Ordered read, length 1–512 bytes
Uses timed shared reader
No dirty-tile inspection shortcut

final tile stores acknowledged
copy owner; invalidate tags

timed color copy
cache miss only
timed upload
fetch prepared work
attachment transfers

Command retirement
Release front packet bank
Advance ordered command stream

Resource alias, not another RAM
1 × SDPB 1024×16
Texture cache OR copy-row scratch
Mutually exclusive owners

timed read

Shared memory subsystem — six bounded client ports, one physical SDR data bus

Copy client
Source reads / masked destination writes

Texture client
External cache refill reads

Scanout client
Current front-buffer reads

Host client
Heap upload writes

Command client
Descriptor / table / reference / record
Diagnostic READ_MEMORY

Tile client
Attachment boot writes
Color / depth loads and stores

Per-client ownership / backpressure
1 request or unconsumed response slot
1–16 × 32-bit words, ≤64 bytes
2 × SDPB 1024×16 per client: 12 banks
Shared write-request / read-response data
Word-fill + synchronous read ports
Held response keeps its slot occupied

Initialization and mandatory refresh
≥100 µs wait → PRECHARGE →
2 refreshes → MODE / timing waits
Refresh deadline ≤15.625 µs
≤1575 cycles at 100.8 MHz
Worst-case request / precharge guard

reserve service before deadline
queued request / response

Request arbitration
Scanout priority quota ≤4 requests
when other clients are waiting
Other five clients round-robin
One active request runs to completion

grant / registered completion

Clocked SDRAM controller / native pins
Open rows / bank selection / ACTIVATE
PRECHARGE / CAS / write recovery
Read↔write turnaround / DQM masks
Sequential native BL8 (32-byte bursts)
Short / unaligned requests still pay BL8

command / address / x32 DQ beats

Embedded SDR SDRAM: 8 MiB, x32
4 banks ×2048 rows ×256 columns
403.2 MB/s raw peak at 100.8 MHz
Word mapping: row11 | bank2 | col8
2 × RGB565 color + shared D24/U32
Heap: textures / records / descriptors / bins

request / response

established vertical-blank pulse
swap only at vblank
per-edge observations

Independent scanout — continues while the ordered executor renders or waits

Free-running display timing
H: 640 +16 front +96 sync +48 back
V: 480 +10 front +2 sync +33 back
Startup stays black until valid memory

pixel enable / H / V

Front / back ownership
Front is read-only to draws / copies
Old front becomes back after swap

current physical color ID

Front-buffer prefetch
1 outstanding tagged SDRAM read
Owned slot: synchronous 32-bit reads
One 32-bit hold → two RGB565 writes
Stale responses discarded at swap

one RGB565 write / edge; slot-read bubbles

RGB565 display FIFO: 2048 pixels
2 × SDPB 1024×16
Synchronous read → later capture
Smaller limits are stress settings

old registered output

Pixel consumer: core / 4 =25.2 MHz
640×480 active, 800×525 total
60 Hz, negative H/V sync
Underflow: black rest of frame
Resynchronize at vertical blank

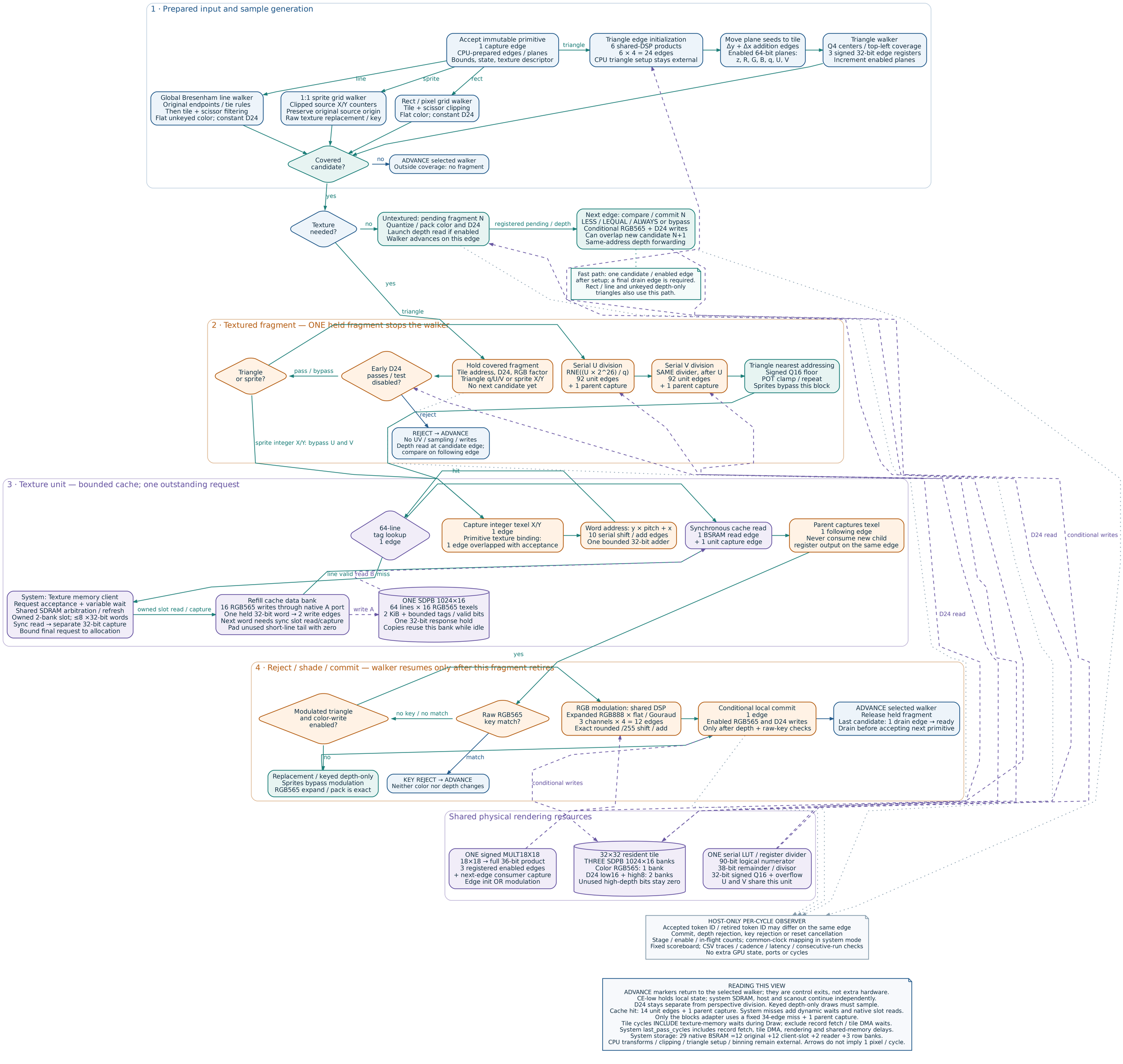
captured completed scanout frame

Host-only observer
SDL / S-key screenshot show actual FIFO output
Raw dumps inspect completed framebuffer
Observer images are not GPU memory

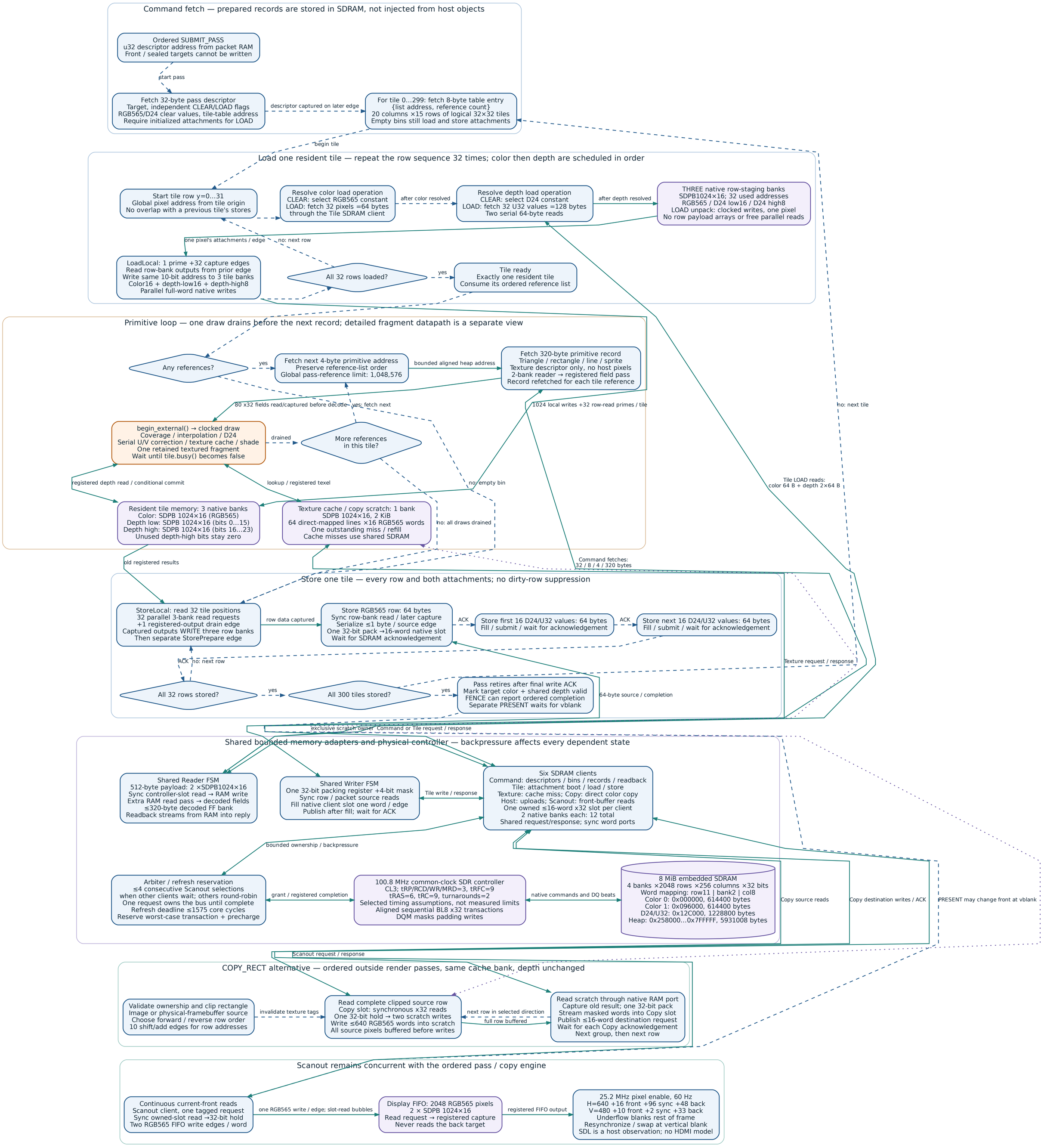
Host-only cycle observer / throughput scoreboard
Distinct accept and retire IDs; stage / enable / occupancy
Common-clock CSV traces and cadence / latency histograms
Adds no GPU registers, queues or ports

Limits that matter
Serial rendering / single active command; no 1-pixel-per-cycle textured pipeline
CPU wide validation checks are simulation assertions, not timed GPU units
29 native BSRAM =12 original +12 client slots +2 reader +3 row staging
Small fields / masks / arithmetic mapped to bounded registers or logic ROM
Whole-system LUT / FF fit and achieved clock remain unproven
Solid teal: data / request-response. Dashed blue: control. Dotted purple: shared resource.
HDMI/TMDS signaling excluded.

Current renderer — streamed storage + throughput instrumentation
Dependency / control-flow view: textured fragments are serialized



Current GPU model — streamed storage + throughput instrumentation
N20K-SYS-0.1 — exact tile lifecycle, SDRAM traffic, copy reuse and scanout



Current traffic / storage limits
Every pass stores all 300 tiles: 1843200 useful color+depth bytes
LOAD of both attachments adds another 1843200 bytes; CLEAR avoids those reads
Boot explicitly writes both color buffers + shared depth: 2457600 bytes
29 native BSRAM =12 original +12 client slots +2 reader +3 row staging
Bulk storage mapped; bounded registers / logic ROM for small fields and arithmetic
LUT/FF physical fit and timing remain unproven; no synthesized clock claim
One active ordered executor operation; only scanout and host parsing progress independently
Host-only per-edge accept/retire observer adds no device state or memory ports
CPU prepared-record consistency checks are simulation assertions, not timed GPU validation units
Solid teal: data / request-response. Dashed blue: control. Dotted purple: reused memory.